

PLUMBING SPECIFICATIONS

- 1.01 SCOPE:**
THESE SPECIFICATIONS TOGETHER WITH THE ACCOMPANYING PLUMBING DRAWINGS ARE INTENDED TO PROVIDE COMPLETE PLUMBING INSTALLATION FOR THE NEW BUILDING AND SHALL INCLUDE ALL LABOR, EQUIPMENT AND MATERIALS.
- 1.02 GENERAL:**
ALL WORK SHALL BE PERFORMED BY SKILLED AND CAPABLE WORKMEN UNDER COMPETENT SUPERVISION, EMPLOYING LATEST AND BEST PRACTICES OF THE TRADE. WORK SHALL BE INSTALLED ACCORDING TO THE ADOPTED LOCAL PLUMBING CODE, AND SHALL MEET WITH PLUMBING INSPECTOR'S APPROVAL IN EVERY RESPECT. LOCAL CODE SHALL APPLY WHERE SUCH CODE EXCEEDS REQUIREMENTS OF THIS SPECIFICATION. IN ABSENCE OF CODE OR AUTHORITIES, INSTALL ALL WORK ACCORDING TO THE 2018 INTERNATIONAL PLUMBING CODE.
- PLUMBING CONTRACTOR SHALL OBTAIN ALL PERMITS AND LICENSES, AT HIS OWN EXPENSE, AND SHALL PAY ALL SERVICE CHARGES REQUIRED FOR PROSECUTION OF THIS WORK.
- PLUMBING DRAWINGS ARE DIAGRAMMATIC ONLY, AND DO NOT SHOW ALL OFFSETS, FITTINGS, ETC. COORDINATE WORK WITH OTHER TRADES, FURNISHING AND INSTALLING ALL FITTINGS, OFFSETS, ETC., REQUIRED AT NO ADDITIONAL COST TO OWNER.
- 2.01 MATERIALS:**
• SOIL, WASTE & VENT PIPING: CAST IRON WITH FOUR BAND NO-HUB CONNECTIONS.
• WATER PIPING: TYPE 1/2" BELOW FLOOR SLAB TYPE 1/2" COPPER ABOVE SLAB.
• PIPE INSULATION: OWENS CORNING AS-1-25 WITH JACKET OR ARMAFLEX CLOSED CELL WITH GLUED JOINTS.
- 2.02 PIPE HANGERS AND SUPPORTS:**
PIPING SHALL BE INSTALLED WITHOUT UNDUE STRESS OR STRAIN ON JOINTS AND EQUIPMENT. HANGERS SHALL BE SECURELY ANCHORED TO BUILDING STRUCTURE. PIPE HANGERS SHALL BE INSTALLED AROUND THE INSULATION WHERE PIPES ARE INSULATED. INSTALL HANGER WITH SHEET METAL SADDLES TO PROTECT THE PIPE INSULATION TO KEEP THE INSULATION FROM CRUSHING.
- MAKESHIFT, FIELD DEVISED METHODS OF PLUMBING PIPE SUPPORT, SUCH AS WITH THE USE OF SCRAP FRAMING MATERIALS, ARE NOT ALLOWED. SUPPORT AND POSITIONING OF PIPING SHALL BE BY MEANS OF ENGINEERED METHODS THAT COMPLY WITH APPROPRIATE PS 42-96. THESE SHALL BE HUBBARD ENTERPRISES/HOLDRITE SUPPORT SYSTEMS OR ENGINEER-APPROVED EQUIVALENT.
- 2.03 FIXTURES:**
ALL FIXTURES SHALL BE NEW, FIRST QUALITY, AND FREE FROM DEFECTS. FIXTURES SHALL BE FURNISHED COMPLETE WITH SUPPLY PIPES, STOP VALVES, TRAPS, FAUCETS, ESCUTCHEONS, HANGERS, SUPPORTS, ETC. ALL EXPOSED PIPING SHALL BE CHROME PLATED.
- WHERE FIXTURES ARE INSTALLED IN CONTACT WITH WALLS OR FLOORS, SEAL THE SPACES AT THE OUTER EDGES OF FIXTURES IN CONTACT WITH WALLS OR FLOORS USING A NON-HARDENING BATHTUB CAULK, 'SELASTIC' BY DOW-CORNING, OR APPROVED EQUAL.
- ALL WALL MOUNTED FIXTURES SHALL BE EITHER MOUNTED ON HEAVY DUTY CONCEALED CARRIERS, HEAVY DUTY WALL MOUNTING BRACKETS WITH THRU WALL BOLTS AND BACK PLATES, OR HEAVY DUTY BRACKETS MOUNTED DIRECTLY TO CONCRETE-FILLED BLOCK WORK WITH STRUCTURAL FASTENERS OF THE 'RED-HEAD' TYPE FASTENED INTO THE CONCRETE FILL. STANDARD LIGHT-WEIGHT PRESSED STEEL MOUNTING BRACKETS WITH SCREWS AND ORDINARY SHIELDS INTO THE SURFACE OF THE BLOCK WILL NOT BE ACCEPTABLE.
- 3.01 CLEANING, PAINTING, AND ADJUSTING:**
AT THE COMPLETION OF THE WORK, ALL PARTS OF THE INSTALLATION SHALL BE THOROUGHLY CLEANED. ALL EQUIPMENT, PIPE, VALVES, AND FITTINGS SHALL BE CLEANED OF ALL GREASE, METAL CUTTINGS, AND SLUDGE WHICH MAY HAVE ACCUMULATED BY OPERATION OF THE SYSTEM FOR TESTING. ANY STOPPAGE, DISCOLORATION, OR OTHER DAMAGE TO PARTS OF THE BUILDING, ITS FINISH OR FURNISHINGS, DUE TO THE CONTRACTOR'S FAILURE TO PROPERLY CLEAN THE PIPING SYSTEM, SHALL BE REPAIRED BY THE PLUMBING CONTRACTOR WITHOUT COST TO THE OWNER. ALL FLUSH VALVES AND OTHER PARTS OF THE SYSTEM SHALL BE ADJUSTED FOR QUIET AND PROPER OPERATION.
- FIXTURES SHALL BE TESTED FOR SOUNDNESS, STABILITY OF SUPPORT, AND SATISFACTORY OPERATION OF ALL COMPONENT PARTS.
- 3.02 INSTRUCTION BOOKLETS:**
CONTRACTOR SHALL FURNISH THE OWNER TWO COMPLETE SETS OF INSTRUCTION BOOKLETS REGARDING THE OPERATION AND MAINTENANCE OF ALL PLUMBING ITEMS OF EQUIPMENT INSTALLED UNDER THIS CONTRACT. BOOKLETS SHALL INCLUDE A COMPLETE PARTS LIST AND TECHNICAL DATA, INCLUDING PREVENTATIVE MAINTENANCE INSTRUCTIONS FOR ALL ITEMS OF EQUIPMENT.
- EACH SET OF INSTRUCTION BOOKLETS SHALL BE NEATLY BOUND INTO A SINGLE UNIT AND PRESENTED TO THE OWNER PRIOR TO FINAL ACCEPTANCE OF THE JOB.
- 3.03 GUARANTEES AND WARRANTIES:**
CONTRACTOR SHALL SERVICE AND MAINTAIN ALL EQUIPMENT INSTALLED BY HIM UNDER THIS CONTRACT FOR A LIKE PERIOD OF 12 MONTHS FROM THE DATE THE CERTIFICATE OF SUBSTANTIAL COMPLETION IS ISSUED, PERFORMING ALL REQUIRED SEASONAL MAINTENANCE.
- CONTRACTOR SHALL GUARANTEE MECHANICAL SYSTEMS AS INSTALLED BY HIM TO OPERATE QUIETLY, SAFELY, AND EFFICIENTLY.



MIDLANDS TECHNICAL COLLEGE ACADEMIC CENTER BATHROOM RENOVATIONS AIRPORT CAMPUS

PROJECT TITLE

OSE # - H59-N178-CL



FELKEL & HASTINGS
Mechanical Engineers
2725 Cypress Street
Columbia, SC 29205
Comm. No.: 22-40c Date: 9-15-22

PLUMBING RENOVATION NOTES

- CONTRACTOR SHALL REMOVE WALL-HUNG WATER CLOSET. FLUSH VALVE, BOLTS, AND COUPLING. CAP WATER LINE IN WALL.
- CONTRACTOR SHALL ADJUST LOCATION OF CARRIER & WATER LINE SO THAT WATER CLOSET IS 18" OFF NEAREST FINISHED SIDE WALL. INSTALL NEW WATER CLOSET, FLUSH VALVE, AND SEAT FOR COMPLETE ADA APPROVED WATER CLOSET.
- CONTRACTOR SHALL DEMOLISH EXISTING URINAL AND MODIFY EXISTING SANITARY SEWER AS REQUIRED TO INSTALL NEW CARRIER & WATER LINE TO FLUSH VALVE AND INSTALL WALL-HUNG CLOSET COMPLETE.
- PLUMBING CONTRACTOR SHALL REMOVE EXISTING URINAL, MODIFY EXISTING WATER AND WASTE PIPING WITHIN THE WALL AS REQUIRED TO LOWER THE NEW URINAL TO 17" ABOVE FINISHED FLOOR. GENERAL CONTRACTOR SHALL PATCH WALL AS REQUIRED.
- PLUMBING CONTRACTOR SHALL REMOVE EXISTING URINAL, MODIFY EXISTING WATER AND WASTE PIPING WITHIN THE WALL AS REQUIRED TO LOWER THE NEW URINAL TO 24" ABOVE FINISHED FLOOR. GENERAL CONTRACTOR SHALL PATCH WALL AS REQUIRED.
- CONTRACTOR SHALL REMOVE LAVATORY, TRAP, AND SUPPLIES. INSTALL ALL NEW FITTINGS ON WATER AND WASTE AS REQUIRED TO TIE BACK INTO EXISTING WASTE CONNECTION TO NEW LAVATORY AND FAUCET.
- CONTRACTOR SHALL REMOVE LAVATORY, TRAP, AND SUPPLIES. INSTALL ALL NEW SANITARY TEE FITTING TO SERVE TWO LAVATORIES, MODIFY EXISTING HOT AND COLD WATER LINES FOR ADDED LAVATORY. INSTALL NEW LAVATORIES COMPLETE.
- CONTRACTOR SHALL INSTALL NEW LAVATORY AND RE-USE EXISTING FAUCET. WHERE LAVATORY HAVE BEEN ADDED, PLUMBER SHALL FURNISH NEW LAVATORY, FAUCET, NEW INSULATED P-TRAP, SUPPLIES AND WIRING TO EXISTING TRANSFORMER.
- EXPOSED FLOOR DRAIN COVER SHALL BE POLISHED WITH STEEL WOOL.
- PLUMBING CONTRACTOR SHALL REMOVE EXISTING WATER CLOSET AND REPLACE WITH NEW SEAL, BOLTS, WATER CLOSET, SEAT AND FLUSH VALVE. REPAIR/CARRIER AS NEEDED.

- THIS DRAWING IS THE PROPERTY AND RECONRANG DESIGN P.A. AND IS NOT TO BE REPRODUCED OR COPIED IN WHOLE OR IN PART. IT IS NOT TO BE USED ON ANY OTHER PROJECT AND IS TO BE RETURNED ON REQUEST.
- MATERIALS, DIMENSIONS AND ALL OTHER CONDITIONS WHICH ARE NOT OTHERWISE NOTED ON THIS DRAWING SHALL BE CONSIDERED AS SHOWN THE SAME UNLESS AS OTHERWISE INDICATED CONDITIONS WHICH ARE MORE FULLY DEFINED ELSEWHERE ON THIS PROJECT OR OTHER DRAWINGS OF THIS PROJECT.
- DO NOT SCALE OFF DIMENSIONS.

NO.	DATE	DESCRIPTION

CD
PROJECT PHASE

2217

BOOMERANG DESIGN PROJECT NUMBER

9.23.2022

DRAWING RELEASE DATE

ENLARGED TOILET ROOM PLUMBING PLAN, DETAILS, AND SPECIFICATION

SHEET TITLE

P102

SHEET

FU COUNT WRONG —
FLUSH VALVE = 6 FU
NOT 4 FU (TANK TYPE)

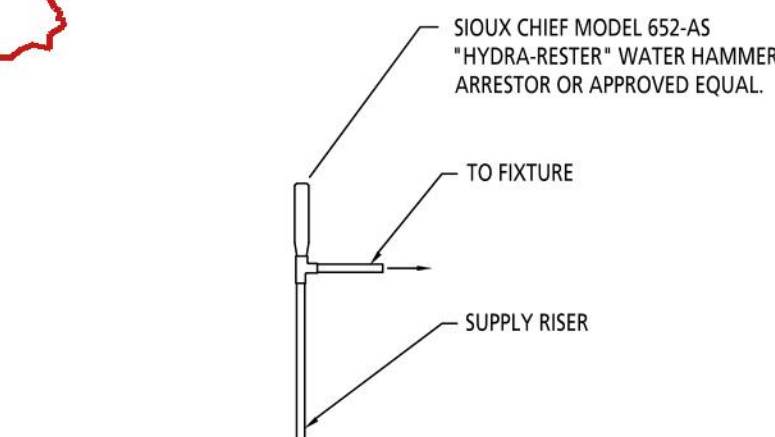
VENT STACK LOC.
CONFLICTS W/ STRUCT.
COORD. W/ ENGR.

3" SAN UNDERSIZED —
FU COUNT REQ'S 4"
MIN — FIX PIPE SIZE

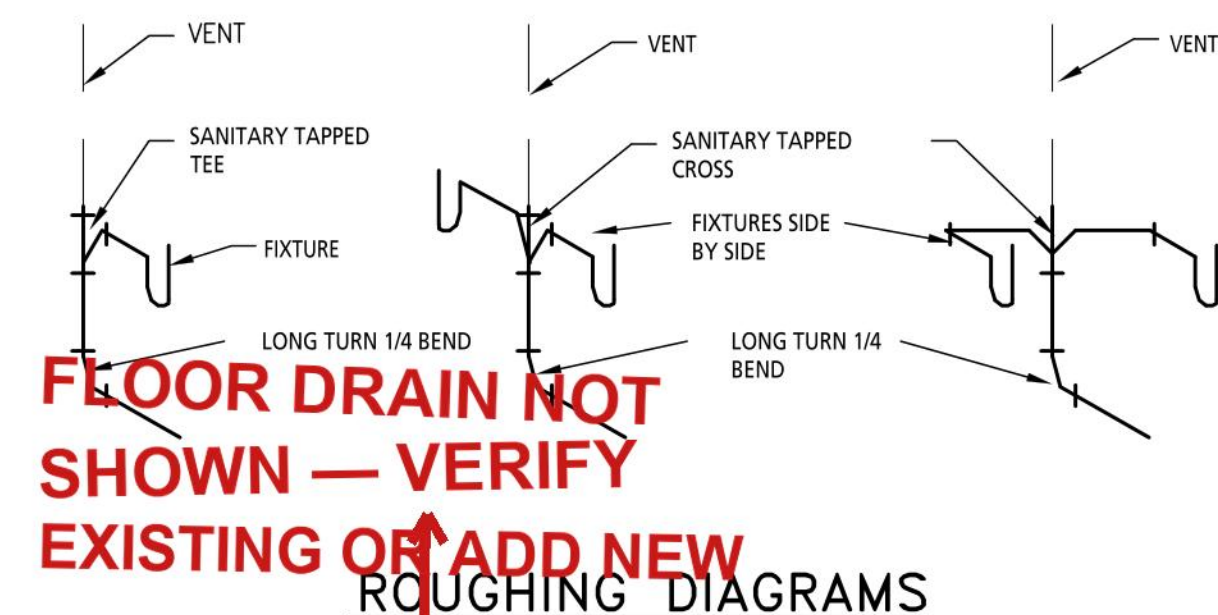
CLEANOUT MISSING
AT DIRECTION
CHANGE — ADD CO

H/C LABELS SWAPPED
AT LAVATORY —
VERIFY AND CORRECT

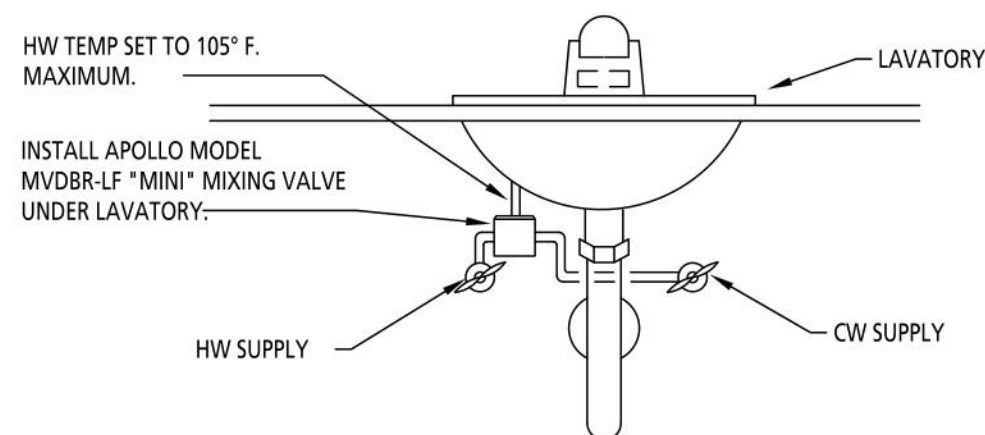
verify?



WATER HAMMER ARRESTOR DETAIL



FLOOR DRAIN NOT
SHOWN — VERIFY
EXISTING OR ADD NEW
ROUGHING DIAGRAMS



P-3 LAVATORY

HOT WATER TEMPERING VALVE DETAIL
NOT TO SCALE

1 THIRD FLOOR GROUP TOILET ENLARGED RENOVATION PLAN 1/2" = 1'-0"